

ORAVEN FINAL PROJECT REPORT



MLB SHORTSTOP
REPLACEMENT
FINDER

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01

EXECUTIVE SUMMARY

Major League Baseball teams often need to identify a realistic shortstop replacement when a player becomes unavailable due to injury, trade, free agency, or age-related decline. The standard approach to this problem relies heavily on summary metrics such as Wins Above Replacement (WAR) and traditional statistics. These tools are useful, but they tend to turn the problem into a search for the best available player rather than the best practical fit. This can cause executives to overlook how a player creates value, what risks come with acquiring that player, and how that player fits within the rest of the roster.

The Oraven project addresses this gap by building a decision-support tool that ranks realistic shortstop replacements using a combination of player archetypes, similarity in playing style, and a structured risk framework. Instead of asking who the best available shortstop is in the abstract, our tool asks who the best practical replacement is given the player being replaced, the type of value that player provides, and the real-world constraints of age, injury history, and contract.

We trained the model on shortstop player-seasons from 2023 through 2025, using Baseball Savant and Statcast for performance data and FanGraphs for injury and contract information. We started in 2023 because several advanced bat-tracking metrics first became available in that season, and those metrics were important for capturing modern offensive profiles. The final tool runs on the 2025 player pool so that the recommendations stay relevant for the current season. After a multistage feature-reduction process that brought the original dataset from more than 200 variables down to a final set of 15, we tested four clustering methods and selected spectral clustering with five clusters using Feature Set D. Feature Set D groups the 15 variables into four interpretable dimensions: power, contact, speed/baserunning, and defense. It produced five archetypes: Elite, Speed/Baserunning, Contact,

Power/Defense, and Balanced. These archetypes align well with how scouts and executives often describe shortstop profiles.

On top of the clustering model, we added a composite risk score that captures three parts of replacement feasibility: age risk, injury risk, and contract risk. We weighted them at 30%, 40%, and 30%, respectively. Injury risk received the highest weight because availability is the most immediate concern in a replacement decision. The final match score combines archetype similarity at 80% with one minus risk at 20%. This setup keeps the focus on player-type fit while still penalizing replacements that are unrealistic to acquire or unlikely to stay healthy.

The final product is an interactive dashboard built for non-technical users such as general managers, executives, and team analysts. A user selects a current shortstop, reviews that player's archetype and risk profile, and then evaluates a ranked list of replacement candidates. Filters for age, contract level, archetype, injury count, and total risk allow the user to narrow the candidate pool based on team needs. That includes both searching for a near-identical replacement and intentionally looking for a different archetype to rebalance the roster. The result is a tool that brings structured analytical reasoning to a decision that has often been driven by intuition and incomplete summary statistics, while remaining simple enough to be useful for MLB executives.

02

PROBLEM DEFINITION

2.1: The Replacement Problem In Major League Baseball

Roster changes in Major League Baseball happen frequently throughout the season. Players get hurt, contracts expire, performance drops off unexpectedly, and trade deadlines push teams to make quick decisions with incomplete information. For executives, replacing a starting shortstop is a risky decision. The shortstop position requires strong defensive contribution while also demanding offensive production and baserunning value, which means a starting shortstop often contributes across multiple parts of the game. Losing that player creates a major disadvantage for the team, and figuring out who fills that spot has implications that go well beyond a single stat category.

The standard approach to this problem relies heavily on Wins Above Replacement, or WAR, a composite statistic that estimates how many additional wins a player provides over a freely available, replacement-level alternative. WAR is genuinely useful as a summary metric, and it appears frequently in executive

discussions, media coverage, and trade analysis. However, treating WAR as the main criterion for replacement decisions creates several problems. First, WAR collapses very different player profiles into a single number, which makes it hard to distinguish, for example, a power-and-defense shortstop from a contact-and-baserunning shortstop with the same overall value. Second, WAR alone says nothing about whether a player is realistically obtainable, how much that player will cost, or how likely that player is to stay healthy. Third, and most importantly, WAR does not capture how a player creates value, which is often the biggest factor when fitting a new player into an existing roster.

2.2: Why Best Replacement Depends On More Than Production

Two shortstops with similar WAR can have completely different archetypes. One might rely on elite range and a strong arm to stop runs on defense while contributing less on offense. Another might combine real power with average defense. A team built around contact hitting and speed will have very different needs than a team that already has several power-oriented bats but lacks defensive range up the middle. The best replacement therefore depends on the player being replaced, the makeup of the surrounding roster, and the strategic direction of the team.

Some constraints make the decision even harder. An ideal replacement might be locked into a long-term contract that makes the player unrealistic to acquire. A player might also have an injury history that makes future availability uncertain. A young player with elite tools might not yet have the track record to support a major acquisition. These constraints are part of the real decision space that executives must also consider.

2.3: A Structured Definition Of A Good Replacement

The goal of the Oraven project was to provide a structured view of comparable player types and alternative archetypes. This allows MLB executives to evaluate both similar players and strategically different options depending on roster needs and constraints. Instead of ranking shortstops on a single axis of overall value, our framework treats replacement as a multidimensional matching problem. In this project, we define a good replacement as a player who either closely matches the original player's profile or provides a similar level of contribution through a different archetype, while remaining feasible in terms of age, injury risk, and contract burden.

03

PROJECT SCOPE AND DATA SOURCES

In this section, we narrow the broad question raised in Section 3 to the specific decision problem addressed in our project and describe the data sources used to build the model. Due to time constraints, we decided to focus on shortstops in this project. Focusing on a single position makes intuitive sense, since teams are most likely to replace a player with another player at the same position. It also reduces the difficulty of categorizing players, since we are comparing like with like. We chose shortstops specifically because the position includes a wide variety of player styles. As arguably the most important defensive position on the field, shortstop is one of the few positions where strong defensive ability can justify lower offensive production. As one example, the top defensive shortstop in our 2025 sample, Bobby Witt Jr., had a fielding run value of +20, while the top defensive first baseman, Matt Olson, had a value of +8. This suggests that defensive ability carries more weight for shortstops than for first basemen and can offset weaker offensive production to a greater degree. Moreover, shortstops can also differ substantially in baserunning ability, which adds another important dimension of variation across the position.

As mentioned before, our project aims to go beyond simple production measures and better capture players' underlying playing styles. For that reason, it made more sense to use detailed Statcast data in addition to traditional counting statistics. This allows us to better understand the underlying performance traits and tendencies of each player than relying solely on traditional statistics. However, because many of the Statcast metrics used in this project are only available beginning in the 2023 season, we were limited to data from the 2023–2025 seasons. This does not directly limit the final recommendations, since the tool only recommends players from the most recent season, 2025, but it does reduce the size of the training sample used for archetype clustering. To address this limitation, we treated each player-season as a separate observation rather than collapsing all seasons for a player into a single datapoint. This substantially increased the number of observations and also provided a useful sanity check, since the same player should generally have the same, or at least a similar, playing style across seasons.

Our project primarily used data from Baseball Savant and FanGraphs. From Baseball Savant, we collected multiple leaderboards containing player Statcast performance metrics across seasons. We then combined those leaderboards into a single master dataset as the first iteration of our modeling file. We also retrieved additional data from FanGraphs on player contracts and injury histories. For contracts, we mainly focused on current average annual value and years left on a contract; we also retrieved player WAR as a performance measure used to

contextualize contract burden. For injuries, we collected injured-list information and injury descriptions. Unfortunately, we did not have access to exact injury and recovery dates. However, we believe that injured-list stints provide a reasonable approximation of injury burden, especially after cleaning outliers. All these data sources can be found in section 15.

In short, our project focuses on identifying replacement options for MLB shortstops using integrated performance, injury, and contract data from Baseball Savant and FanGraphs.

04

DATA PREPARATION AND FEATURE ENGINEERING

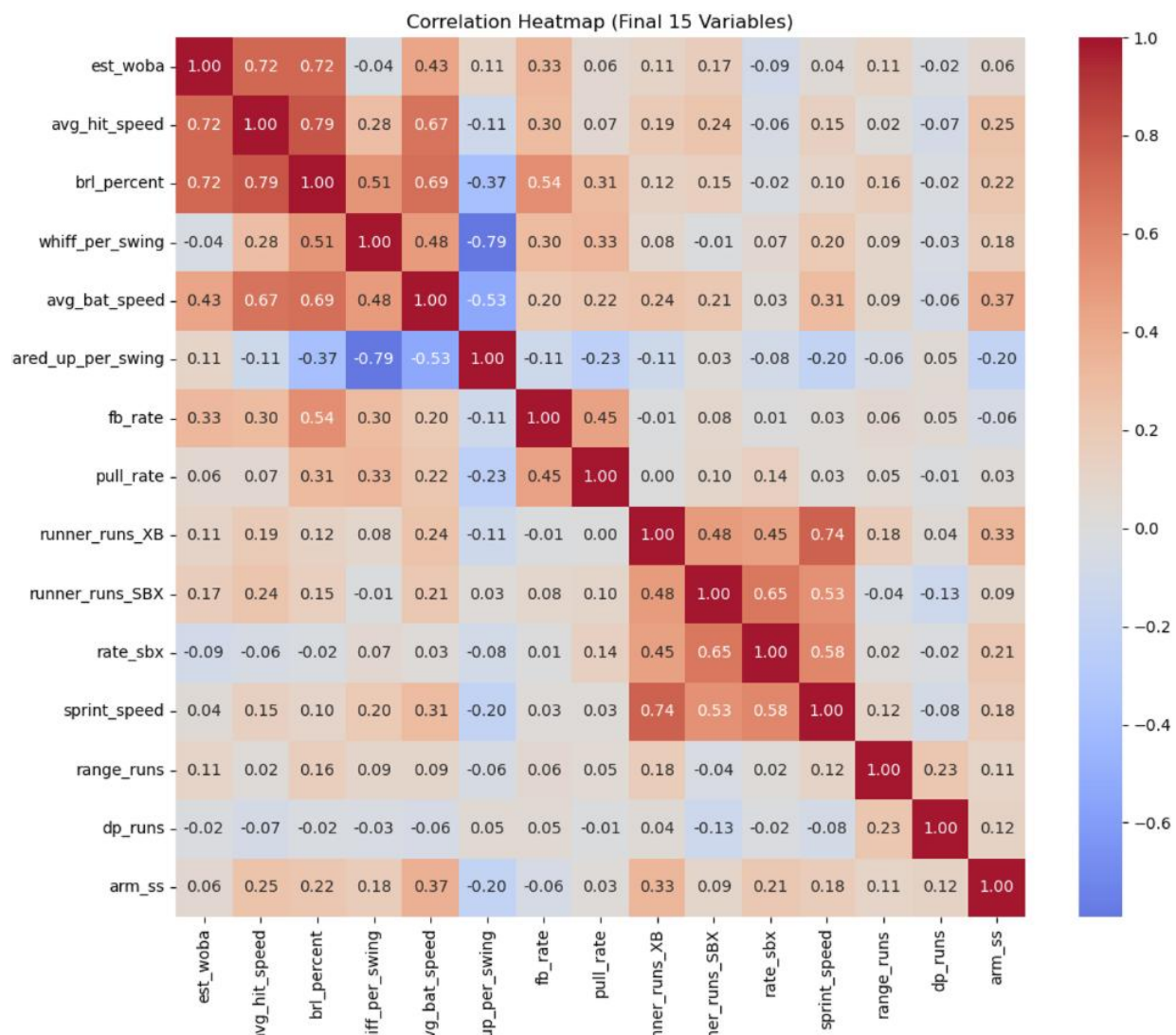
Our initial dataset was built by combining multiple Statcast leaderboards from Baseball Savant covering offensive, defensive, and baserunning performance. We selected 2023 through 2025 as our training window because several advanced batting metrics, such as squared-up percentage, only became available beginning in 2023. The raw dataset contained over 200 variables at the player-season level, which was too large for reliable clustering. As a result, we used a correlation-based screening process to reduce the number of variables.

This reduction process had four main stages. First, we removed direct duplicates and low-variance variables, which reduced the feature count to 181. Next, we ranked the remaining variables into three categories based on expected usefulness: highly likely, potentially useful, and unlikely. After dropping variables in the unlikely category, we grouped the remaining variables into naturally related categories such as bat tracking, exit velocity, defense, and baserunning. We then ran correlation checks within each group and removed redundant variables, which brought the count down to roughly 25 variables. Lastly, we used PCA, domain knowledge, and discussions of variable importance to determine which variables to retain, arriving at a final set of 15 variables. This process balanced statistical screening with baseball interpretability, ensuring that the final feature set remained both analytically useful and meaningful in baseball terms. Table 1 organizes the final 15 variables by performance domain.

Table 1. Final 15 variables grouped by performance domain.

- **Offense (8):** average bat speed, squared-up percentage, whiff rate (per swing), average hit speed, barrel percentage, estimated wOBA, fly ball percentage, pull percentage
- **Defense (3):** arm strength, range runs, double play runs
- **Baserunning (4):** sprint speed, extra-base value, stolen-base value, stolen-base success rate

A correlation heatmap of the final 15 variables, shown below, indicates that most pairwise correlations are modest. This supports the idea that the selected variables capture meaningfully different aspects of shortstop performance. This compact and balanced feature set provided the foundation for the clustering analysis described in the next section.

Figure 1. Correlation heatmap of the final 15 variables.

05

WHY WAS CLUSTERING NEEDED

In practice, teams do not replace players based only on statistical value; they replace player roles. Players with similar overall value may still contribute to their teams in very different ways. For example, a speed-oriented shortstop and a power-oriented shortstop may have similar overall value but very different team fit. Clustering allowed us to move from “similar production” to “similar archetype,” making the replacement system more realistic and easier to interpret. It also gave the user more flexibility, since the tool can be used either to find a similar archetype or to intentionally choose a different fit depending on team needs.

06

CLUSTERING METHODOLOGY

We evaluated several clustering methods across four modified feature sets and multiple values of k , the number of clusters. In the end, we selected spectral clustering with Feature Set D and $k=5$, which we refer to as **5D**, as the final model. This section describes that selection process in detail.

6.1: Methods Explored

We explored four clustering methods:

- **K-means:** partitions observations by minimizing within-cluster variance under Euclidean distance.
- **Gaussian Mixture Model (GMM):** a probabilistic clustering approach that models the data as a mixture of Gaussian distributions.
- **K-medoids:** similar to k-means, but uses actual observed data points as cluster centers rather than estimated centroids.
- **Spectral clustering:** projects the data into a low-dimensional embedding defined by the eigenvectors of a graph Laplacian, then applies k-means in that embedded space.

6.2: Evaluation Criteria

We evaluated the candidate models from two main perspectives.

First, we assessed overall clustering performance by asking whether the resulting clusters were compact and well separated. We measured this using the silhouette score and the Dunn index.

Second, we examined the mean values of the variables within each cluster, along with the specific players assigned to those clusters, in order to identify baseball patterns and interpret the groups in practical terms. We did not aim only for strong statistical scores; cluster interpretability was equally important in the final model choice.

By combining internal validation metrics with baseball interpretability, we aimed to identify a model that was both analytically strong and useful for the final recommendation tool.

6.3: Feature Set Strategy

We tested four versions of the final feature space:

- **A:** original 15-variable feature set
- **B:** trimmed version of the original feature set, most commonly removing pull rate and bat speed
- **C:** version that applied PCA to the offensive variables
- **D:** version that grouped the final 15 variables into four composite dimensions:
 - Power
 - Contact
 - Speed/Baserunning
 - Defense

Feature Set D was designed to preserve the major dimensions of shortstop performance while reducing the risk that offense-heavy raw variables would dominate the clustering structure. Because it grouped the variables into interpretable baseball dimensions, it was especially well suited for a final tool meant to be understandable to non-technical users.

6.4: Final Model Selection

We first calculated silhouette scores for each clustering method across the four feature sets and multiple values of k between four and ten. Among the methods tested, only spectral clustering produced consistently strong results overall. We therefore examined the spectral candidates in more detail and calculated the Dunn index for the leading models. Table 2 summarizes the approximate silhouette-score ranges observed across the main clustering methods and shows the clear advantage of spectral clustering over the more classical alternatives.

We ultimately selected spectral clustering 5D, meaning Feature Set D with $k=5$ clusters, as the final model. Although some spectral models built on Feature Sets A, B, and C also performed reasonably well, we did not choose them because their cluster assignments were less consistent with the baseball archetypes we wanted the final tool to represent. Higher- k models often revealed finer structure, but they also became too fragmented for practical use in the final recommendation tool. Additional clustering comparisons and eigenvector-space plots are included in appendix A.

Table 2. Clustering model silhouette scores.

Model	Silhouette Score
K-means	0.10-0.15
Spectral clustering	0.50-0.55
K-medoids	0.07-0.13
GMM	0.12-0.16

07

FINAL SHORTSTOP ARCHETYPES

In this section, we discuss how we interpreted the final clustering results. This is a qualitative analysis based on the mean values of each cluster and the players assigned to them. It is an essential step, because clustering players without being able to interpret the resulting groups would not be useful for making recommendations. Before discussing the individual clusters, it is important to note that they should not be interpreted primarily as differences in average player quality. Instead, each cluster represents a player archetype, within which there may still be stronger and weaker individual players. We assigned the cluster labels by combining the cluster means with the real players contained in each group.

We interpreted the first cluster as the **Elite** cluster. These players tend to be above average in power, baserunning, and defense. Example players from the 2025 season in this cluster include Bobby Witt Jr. and Elly De La Cruz, both of whom stand out for their combination of hitting, running, and defensive ability. This cluster does tend to include players toward the higher end of overall quality, since its members are

above average in three dimensions. However, it should not automatically be viewed as the “best” cluster. Because baserunning is typically valued less than hitting or fielding, some players from other clusters may still produce higher WAR totals than certain players in the Elite cluster.

We interpreted the second cluster as the **Speed/Baserunning** cluster. Players in this cluster share the common trait of being fast and effective on the bases, although they may differ in other dimensions. Example players in this cluster include Trea Turner and CJ Abrams, both of whom recorded more than 30 stolen bases in the 2025 season. However, the two players differ substantially in fielding value, which helps explain why Trea Turner more than doubled CJ Abrams’ WAR total (6.7 versus 3.1). This example highlights that players within the same cluster can still differ considerably in overall quality.

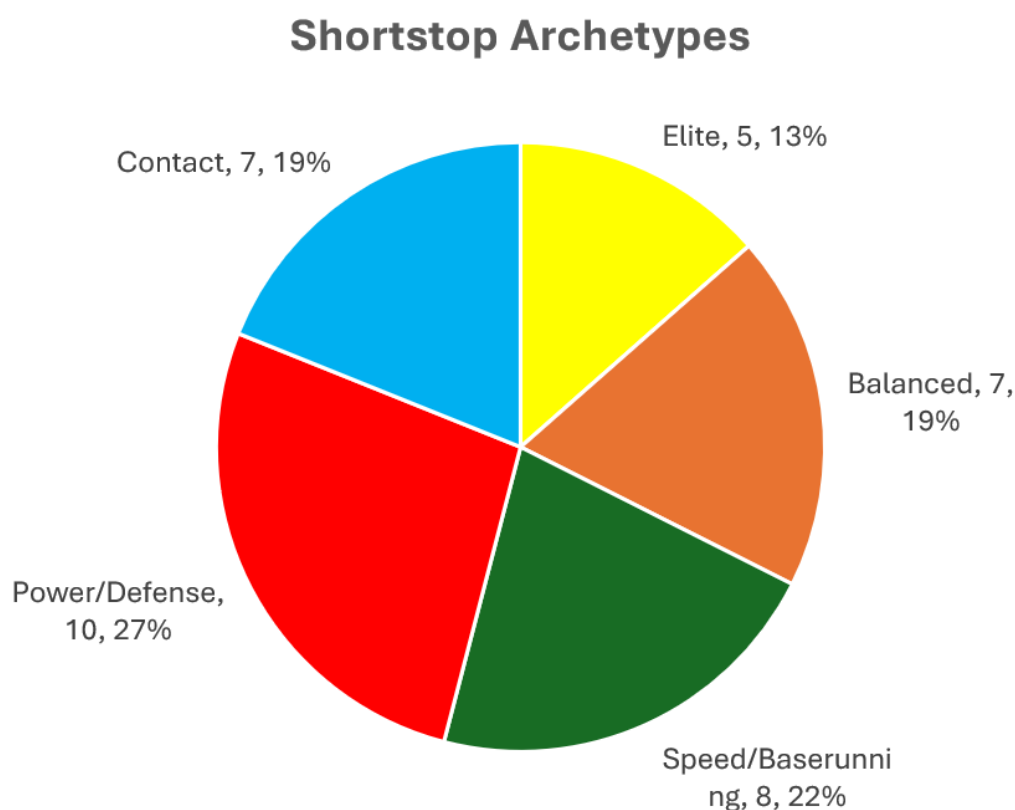
We identified the third cluster as the **Contact** cluster. These players are defined by strong bat-to-ball skills and low strikeout tendencies, although their total value may still differ depending on their power and defensive contributions. Example players in this cluster include Geraldo Perdomo and Joey Ortiz, both of whom posted strikeout rates below 15% in 2025, compared with a league average of about 23%. Yet because of the gap in power between the two players, Perdomo accumulated 7.1 WAR in 2025, compared with 0.7 WAR for Ortiz.

The fourth cluster in our results was labeled **Power/Defense**. These players combine strong defensive performance with power-oriented hitting tendencies. It should be noted that power-hitting tendencies do not always translate directly into elite power production, since contact ability still varies across players in this cluster. Example players in this cluster include Corey Seager and Willy Adames. At first glance, this cluster may seem less intuitive from a traditional baseball perspective, where defense is often associated more closely with athleticism and baserunning than with power hitting. However, many of the players in this cluster are established veterans. Their defensive value may therefore come more from experience, positioning, and anticipation than from raw athleticism alone.

Finally, we labeled the remaining cluster **Balanced**. These players are relatively well rounded, but they do not stand out as extreme in any one dimension. Notable players in this cluster include Bo Bichette and Xander Bogaerts. Both have had stronger seasons in the past, but in 2025 they fit more naturally into a balanced, non-elite profile.

As a final note, the clusters are not separated by wide empty gaps in the spectral embedding space, as seen in appendix B, which reflects the complexity and multidimensionality of the original data. As a result, players near the boundary of a cluster may in some cases be more similar to players in a neighboring cluster than to players at the center of their own group. Even so, we consider the clustering successful because all five groups are interpretable in baseball terms, and the players assigned to each cluster generally fit the meaning of that archetype. Figure 2 below shows the breakdown of the clusters.

Figure 2. Pie chart of archetype breakdown



08

RISK MODEL DESIGN

When MLB front offices make decisions about trading for or signing a player, they cannot rely only on the target's on-field performance. Because of budget limitations and roster constraints, teams generally want to avoid players who are declining, injury-prone, or significantly underperforming relative to their contracts. To reflect this reality in our model, we incorporated a risk-adjustment framework based on age, contract burden, and injury history. In this section, we discuss the design of the risk model and how its output should be interpreted.

8.1: Overall Risk Framework

As discussed above, we treat age, injury history, and contract burden as the three main dimensions of acquisition risk. In theory, these three dimensions may be related, since older players often have larger contracts and longer injury histories. However, because we do not have a well-defined target variable for risk, we could not easily estimate interaction terms among these dimensions using a regression framework. Instead, the model assigns a partial risk score to each dimension and combines them through a weighted sum to produce the total risk score.

The first partial risk factor, age risk, is designed to capture the possibility of decline and concerns about long-term durability through player age. The model does not assign the lowest age risk to players in their prime. Instead, it assigns the lowest age-risk scores to the youngest players, reflecting the belief that they have more room for improvement and less concern about decline.

The second partial risk factor, injury risk, is designed to capture how likely a player is to be injured and miss time in the future by looking at the player's injury history. To capture this properly, we determined that simply counting career injuries was not sufficient. Instead, we incorporated missed days and injury descriptions in order to better represent both frequency and severity.

The third partial risk factor, contract risk, is designed to capture the financial burden and loss of roster flexibility associated with acquiring a player relative to that player's current on-field production. For this purpose, the model should not assign the highest contract risk simply to the players with the largest salaries or the longest remaining terms. Instead, it should assign higher contract risk to players whose financial burden is not matched by their current level of performance. To support this part of the model, we included player WAR as a simple measure of seasonal performance quality.

All three partial risk factors are scaled to the same range, between 0 and 1, so they can be combined directly into a total score. The total risk score is calculated as:

$$\text{Total Risk} = 0.3(\text{Age Risk}) + 0.4(\text{Injury Risk}) + 0.3(\text{Contract Risk})$$

The weights were chosen so that the total risk score remains on the same 0-to-1 scale as the individual partial risks. Injury risk receives the largest weight because an injured player cannot contribute on the field, which we consider the most immediate risk in an acquisition decision.

8.2: Scaling and Normalizing Functions

As noted above, the model uses multiple raw metrics to construct the injury and contract risk scores. We therefore assign sub-risk values to each metric and then combine them into the final injury and contract risk scores. To do this, we use three different scaling functions depending on the type of data being normalized. Our primary scaling function is percentile-based scaling, defined as:

$$\text{Partial Risk}(x) = \frac{\mathbf{1}\{X_i < x\} + 0.5 * \mathbf{1}\{X_i = x\}}{N}$$

This was our primary scaling function because it is robust to outliers and has a simple interpretation as the percentile ranking of a player on a given feature.

Our secondary scaling function is log min-max scaling, defined as:

$$\text{Partial Risk}(x) = \frac{\log(x) - \log(X_{\min})}{\log(X_{\max}) - \log(X_{\min})}$$

This function was used when we wanted to preserve information contained in statistically extreme values rather than compressing them into percentile ranks. It helps retain the scale of differences observed in the original feature space.

The third function is a piecewise constant function. This function was used specifically for age risk, where neither percentile ranking nor min-max scaling made much practical sense.

8.3: Age Risk

The age-risk model uses a piecewise constant function to map player age to a score between 0 and 1, where younger players receive lower risk values. Specifically, the age-risk mapping is shown in Table 3.

Table 3. Age Risk Mapping

Age Range	Age Risk Score
Under 22	0.05
22 – 26	0.10
27 – 29	0.25
30 – 32	0.45
33 – 35	0.70
36+	0.90

One reason we chose such a simple structure instead of fitting a more complex model is the limitation of our dataset. As discussed in the data-sources section, our data only span the 2023 through 2025 seasons. In other words, for each player, we only observe a short segment of the player's career rather than a full aging trajectory. If we had tried to fit a regression model using only this short time window, the result would likely have been skewed toward the age range of the best current players rather than the true long-run aging curve of a shortstop. Even with more complete data, we still believed it was intuitive to assign a lower risk score to younger players because they generally have more room for improvement and less concern about decline. We therefore preferred this approach because of its simplicity and interpretability, making the age-risk component easy to explain even to readers without a technical background.

8.4: Injury Risk

The injury-risk model can be decomposed into three main parts: missed time, injury frequency, and injury severity. Before discussing these individually, it is important to note the main limitation of our injury data. We were only able to obtain Injured List (IL) data rather than exact injury and recovery dates. In other words, we approximated injury duration using the dates on which a player was placed on and removed from the IL.

The first part of the injury-risk model is actual missed days. Actual missed days were approximated by summing the durations of all IL stints. After calculating this quantity for each player, we used log min-max scaling to map it onto the interval from 0 to 1.

A limitation of this approximation is that players are automatically removed from the IL when the season ends, even if the underlying injury would have required more time. To address this issue, we also incorporated expected missed days. The model calculates expected missed days by approximating each injury stint using the average duration of all IL stints with the same injury type. This helps smooth cases where the observed IL stint was cut short by the end of the season. We again used log min-max scaling to map this metric to the 0-to-1 range.

The raw injury-frequency measure is defined as:

$$\text{Injury Frequency} = \text{Number of Injuries} + 2 * \max(\text{Number of Repeating Injuries})$$

We designed the raw frequency measure this way because repeated injuries were viewed as more concerning than distinct injuries, due to the possibility of relapse. Because this measure is typically a small integer, we used percentile-based scaling to map it onto the interval from 0 to 1.

For injury severity, the model takes the longest IL stint for each player as the raw severity measure, then applies percentile-based scaling. We included this metric because a single long IL stint may signal a more severe injury, which could have a larger effect on future durability than a collection of smaller injuries summing to a similar total number of missed days.

The final injury-risk score is calculated as:

$$\text{Injury Risk} = 0.35(\text{Actual Days Risk}) + 0.25(\text{Expected Days Risk}) \\ + 0.25(\text{Injury Frequency}) + 0.15(\text{Injury Severity})$$

The weights reflect our judgment about the relative importance of these different injury dimensions.

One possible concern is that the injury metrics may already reflect age. In theory, older players might be expected to accumulate more injuries or more missed days. However, within our dataset, the correlations between age and injury measures were weak and statistically insignificant. We believe this is partly a consequence of the short 2023–2025 time window. As a result, we did not directly adjust injury risk for age.

A second point worth noting is that all players receive a strictly positive final injury-risk score, even those without an observed injury history. This occurs because

the percentile-based scaling function assigns tied observations the average rank, which still produces a positive value when many players have no injury history. We chose not to force these scores to zero, because even players without prior injuries still face some risk of being injured in the future.

8.5: Contract Risk

The contract data retrieved from FanGraphs contained three key pieces of information: average annual value (AAV), years remaining on the contract, and contract type. As discussed earlier, we also retrieved player WAR by season to evaluate contracts relative to performance.

Among these pieces of information, we ultimately decided not to include contract type directly in the final contract-risk score. In MLB, the major contract types are rookie contracts, arbitration contracts, and free-agent contracts. These categories are already strongly tied to salary level, since rookie contracts are near the league minimum, arbitration contracts act as a bridge toward free-agent salaries, and free-agent contracts typically contain the highest financial values. Because contract type is so closely related to AAV, we chose to use AAV directly in order to avoid overweighting the same information twice.

The AAV risk score uses each player's contract AAV as raw data. Because the dataset contains a wide range of contracts, from low-cost rookie deals to large free-agent contracts, AAV includes meaningful outliers. However, because teams care more about actual dollar value than percentile ranking when evaluating contracts, we used log min-max scaling to preserve those financial differences.

The motivation for using log min-max scaling came from the fact that percentile scaling did not preserve meaningful financial differences. For example, the percentile gap between two rookie contracts could be larger than the percentile gap between a top rookie contract and a mid-level free-agent contract, even though the actual dollar difference in the second case was far greater.

The years-left score uses the number of years remaining on a contract as raw data. Intuitively, contracts with more years remaining are riskier because they create a longer-term financial commitment and reduce roster flexibility. Since contract length is usually a small integer, we used percentile-based scaling for this part of the model.

Finally, we use the player's seasonal WAR as a discount factor. This reflects our belief that more productive players are more deserving of stronger contracts. Since WAR values in a single season generally fall into a manageable range, we used percentile-based scaling to map WAR into a 0-to-1 score.

The final contract-risk score is defined as:

$$\text{Contract Risk} = \max(0, 0.75 * (\text{AAV Risk}) + 0.25 * (\text{Years Left Risk}) - \text{WAR Score})$$

We impose a lower bound of 0 because otherwise a player on a cheap contract with strong performance could receive a negative contract-risk score. In those cases,

the contract-risk score is set to 0, indicating a contract that is effectively risk-free within this framework.

8.6: Interpretation of Total Risk

Because we attempted to spread the risk scores at each stage of the sub-risk construction, the final total-risk distribution across our dataset is roughly bell-shaped on the interval from 0 to 1, with a somewhat heavier left tail. This heavier left tail is largely explained by the fact that many players in our dataset are on rookie contracts and have little or no injury history. See appendix C for a graph of the risk distribution. The simplest interpretation of the total-risk score is that lower is better. Based on the empirical distribution of our scores, we use the rule-of-thumb interpretation shown in Table 4.

Table 4. Interpretation of Total Risk

Score Range	Risk Level	General Meaning
0.00 – 0.20	Low	Young, cheap, healthy — ideal acquisition target
0.20 – 0.50	Moderate	Some risk factors present but manageable
0.50 – 0.70	Elevated	Multiple risk factors — requires careful evaluation
0.70 – 1.00	High	Old, expensive, and/or significant injury history

From the perspective of our recommendation system, players in the “High Risk” category should generally be approached cautiously. At the same time, players in the “Low Risk” category may also be more difficult to acquire, since they are often young, inexpensive, and relatively healthy. In the following section, we explain how risk is integrated into the broader recommendation framework.

09

RECOMMENDATION FRAMEWORK

The final matching percentage used to rank replacement options is defined by the following formula:

$$\text{Match Score} = 0.8 \times \text{Similarity} + 0.2 \times (1 - \text{Risk})$$

The similarity component is calculated as:

$$\text{Similarity} = \frac{1}{1 + \text{Distance}}$$

Here, distance is the Euclidean distance between two players in the spectral embedding space.

The similarity score captures closeness in archetype and player-profile space, while the risk score penalizes players who may be more difficult to rely on or acquire. We chose the 80/20 weighting because player-type fit should drive the replacement decision, while risk should act as a secondary adjustment. As a result, the final match score prioritizes similarity while still preserving realism. This structure ensures that the tool does not simply return the lowest-risk players or the statistically closest players in isolation, but instead balances both considerations.

10

DASHBOARD/FINAL TOOL

The final product of this project is an interactive dashboard that allows users to explore shortstop replacement options in a simple and practical way. The dashboard was designed for non-technical users, such as MLB executives or managers, who may not be familiar with the underlying clustering or risk-modeling methods but still need clear and interpretable outputs. Rather than requiring users to understand the mathematics of the model, the dashboard translates the full recommendation framework into a decision-support interface.

The workflow begins with the player-selection dropdown menu. A user first selects the shortstop they are trying to replace from the 2025 player pool. Once a player is selected, the dashboard displays that player's profile on the left side of the screen. This profile includes the player's archetype label, WAR, age, number of injuries, and average annual value (AAV) of the contract. It also shows the player's

injury risk, contract risk, and total risk, along with a short automatically generated summary of the player profile.

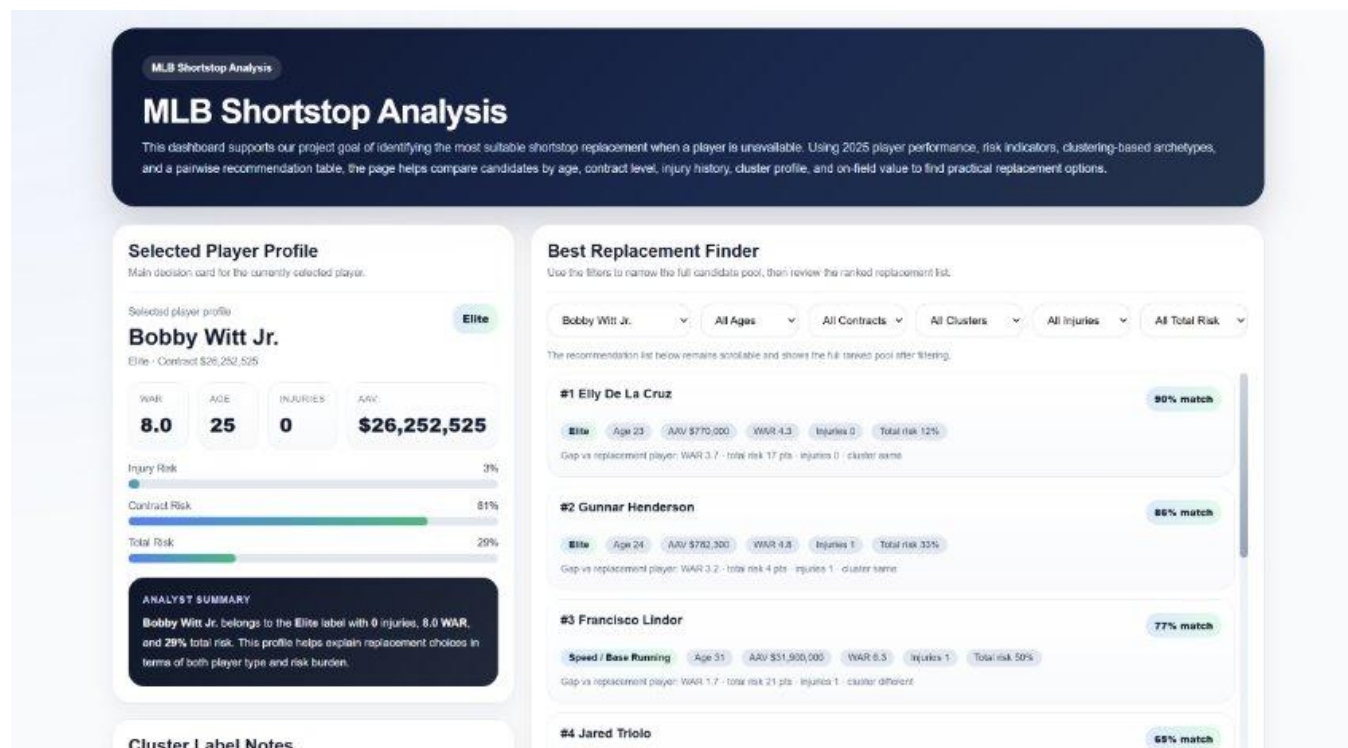
On the right side of the dashboard, the tool returns a ranked list of replacement options. Each recommended replacement is displayed with a match percentage, archetype label, and key information such as age, contract level, injury count, and total risk. The dashboard also reports how the replacement differs from the originally selected player, which helps users quickly understand not only who the recommended players are, but also why they appear in the ranking.

A key feature of the dashboard is the set of filters that appear across the top of the recommendation panel. These filters allow the user to narrow the candidate pool by age, contract level, cluster, injury history, and total risk. This makes the dashboard more flexible than a simple statistical comparison tool. A front office user can either view the full ranking or impose practical constraints to focus on the subset of players that best matches a team's budget, risk tolerance, or roster strategy.

The dashboard is available in the project repository at:

<https://github.com/GregoryHeiger/MLB-Shortstop-Archetypes>

Figure 3. Final dashboard interface for the MLB Shortstop Replacement Finder.



11

EXAMPLE APPLICATIONS

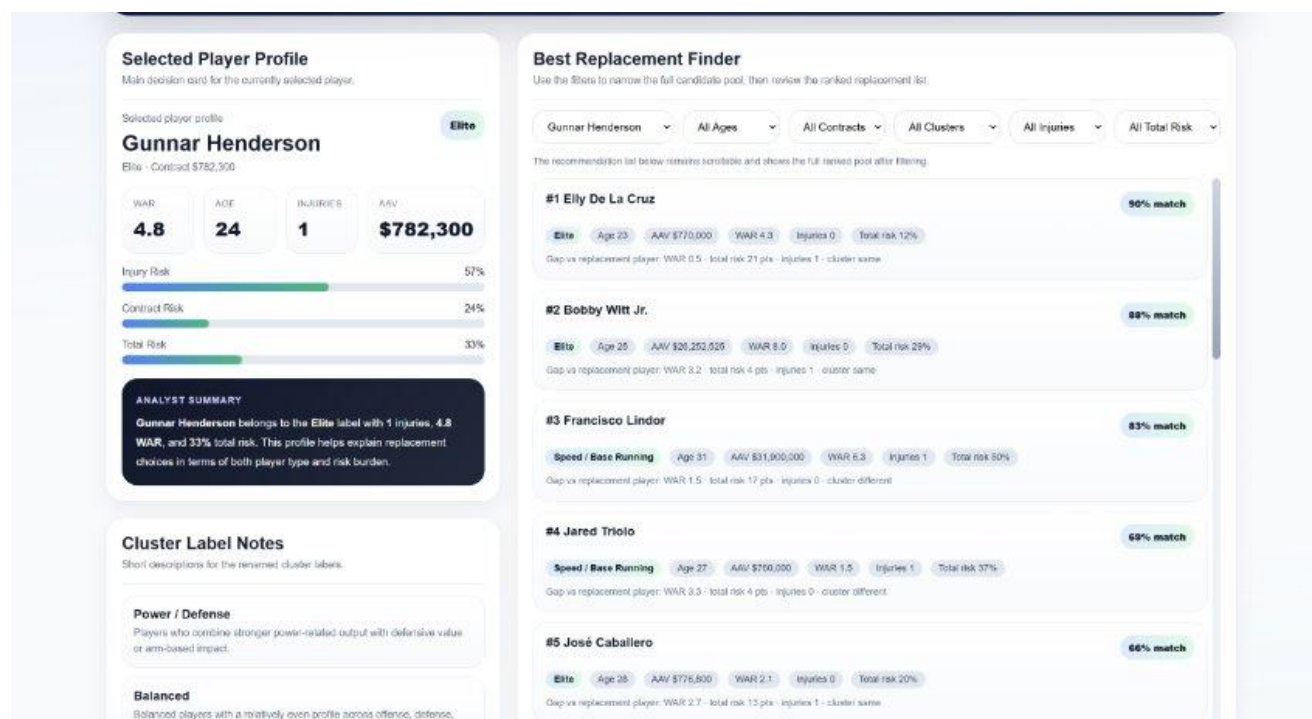
To demonstrate how the dashboard works in practice, we applied the tool to several example players from the 2025 shortstop pool. These examples show that the system does not simply rank the “best” shortstops overall. Instead, the recommended replacements depend on the profile of the player being replaced, the team’s constraints, and the filters selected by the user.

11.1: Replacing an Elite Shortstop

Our first example uses Gunnar Henderson of the Baltimore Orioles. Gunnar appears in the dashboard as a relatively young player on a rookie contract with a low total-risk score and an **Elite** archetype label. Once he is selected, the replacement list is led by players such as Elly De La Cruz and Bobby Witt Jr., who are also elite all-around shortstops.

This result is consistent with the design of the model. When the selected player is an elite, high-impact shortstop, the strongest matches should come from players with similar overall profiles rather than from shortstops who simply have acceptable value. In this case, the dashboard behaves as intended by returning replacements that are close in archetype and overall contribution.

Figure 4. Gunnar Henderson replacement example

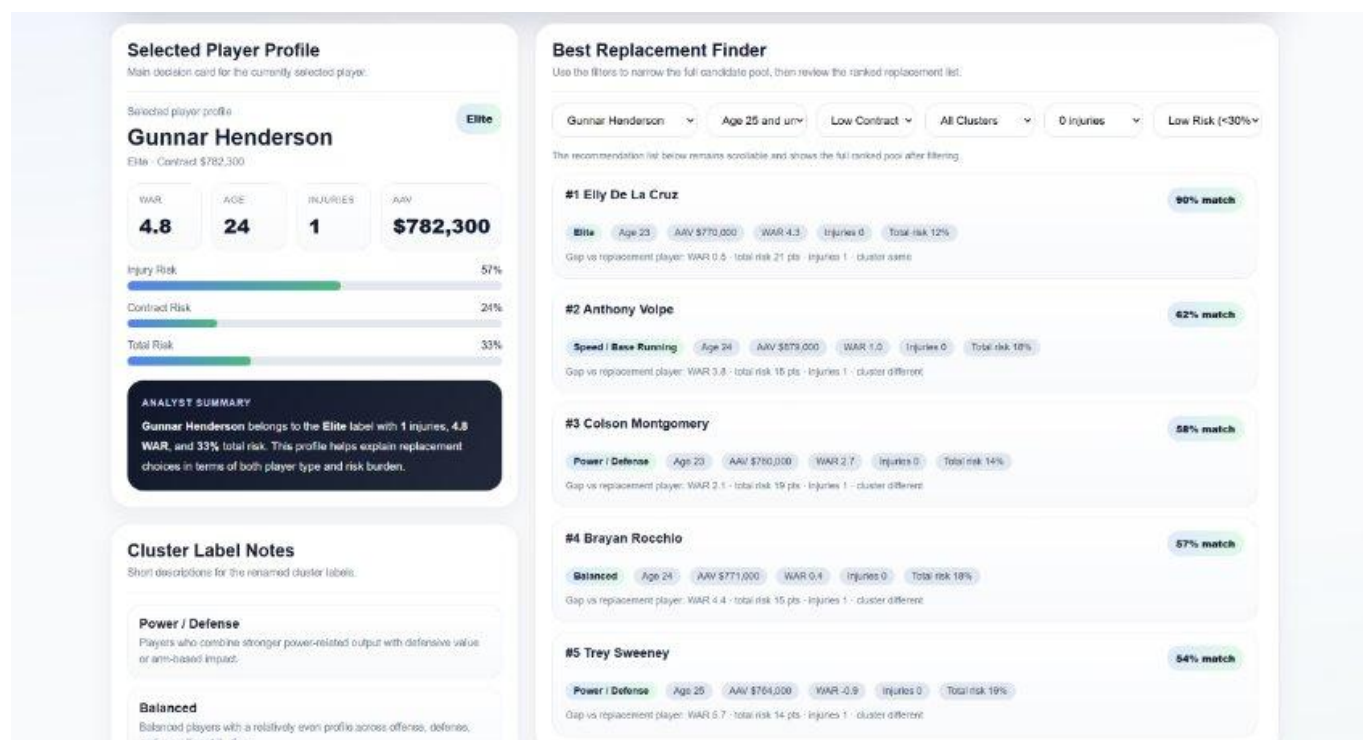


11.2: Filtering the Elite Example

The Gunnar Henderson example also illustrates the strategic value of the dashboard filters. By applying filters for age under 25, low contract burden, zero injuries, and low total risk, the full recommendation list can be narrowed to a smaller, more targeted shortlist. In our test case, these filters reduced the candidate pool to five players out of the 37 shortstops in the 2025 sample.

This feature is important because the top-ranked replacement is not always the most realistic option in a front-office setting. A team may face financial, developmental, or roster constraints that make the full recommendation list too broad. The filter system allows the user to move from a general ranking to a more actionable shortlist based on the organization's priorities.

Figure 5. Gunnar Henderson example with filters applied

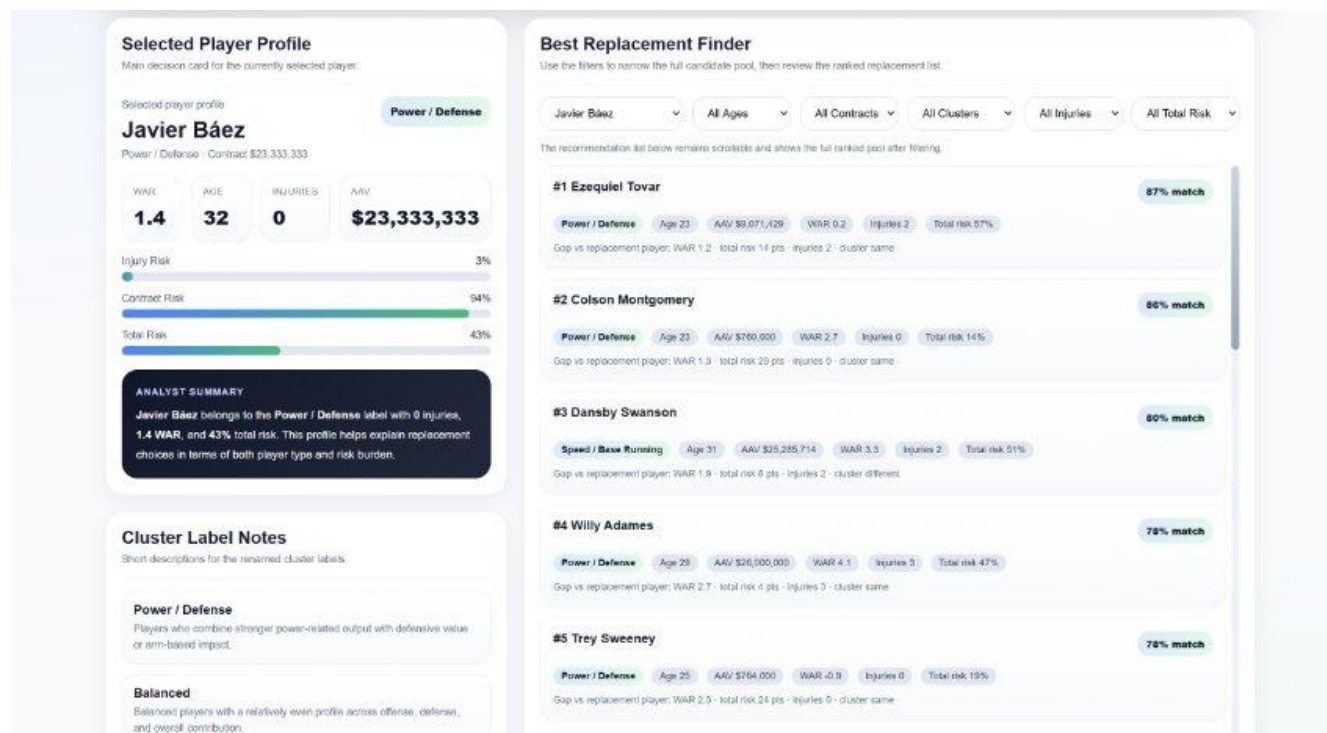


11.3: Replacing a Different Player Profile

Our second example uses Javier Báez, whose 2025 profile fits the **Power/Defense** archetype rather than the **Elite** archetype. In this case, the dashboard produces a different set of recommendations, with top options coming largely from the same power-and-defense-oriented profile instead of from the elite all-around group. This confirms that the system is not simply recommending the highest-value players at the position. Instead, it adjusts to the profile of the player being replaced.

This distinction is central to the project. A replacement decision is not always about maximizing overall value in the abstract; it is often about replacing a specific player type with one who contributes in a similar way.

Figure 6. Javier Báez replacement example

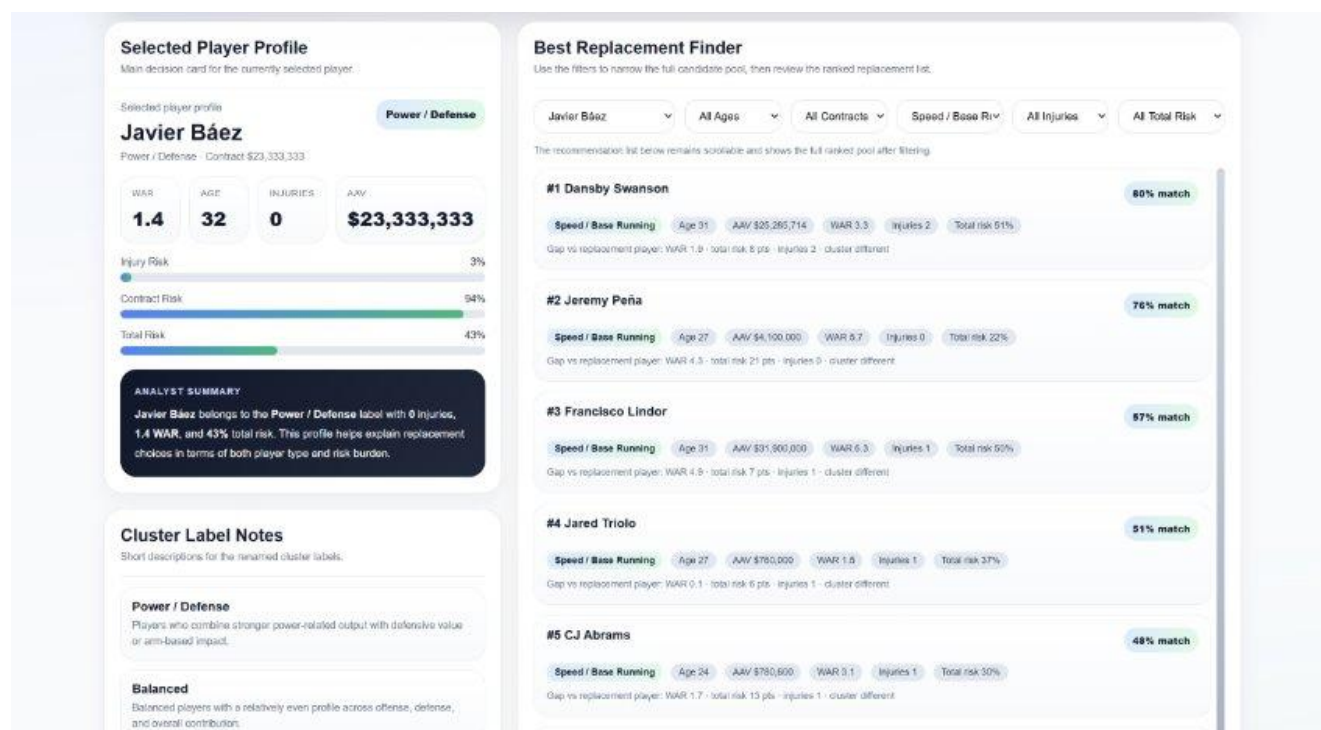


11.4: Intentionally Choosing a Different Fit

The Báez example also highlights another use case: teams may intentionally want a different profile instead of the closest same-archetype replacement. For example, a team that already has several power-oriented hitters might prefer to add more speed or contact to balance the roster. By filtering the candidate pool toward the **Speed/Baserunning** archetype, the dashboard can return a different set of recommendations that better matches that strategic goal. Users can see the different shortstop archetypes on the bottom left of the dashboard.

This feature makes the dashboard more flexible than a traditional similarity model. Users can search either for the closest role match or for a different player type that still provides a realistic replacement option. This ability to move between same-archetype and different-archetype recommendations is one of the main strengths of the final tool.

Figure 7. Javier Báez example with alternate cluster preference



12

CHALLENGES AND ITERATION

Developing the MLB Shortstop Replacement Finder required substantial iteration across the modeling pipeline, recommendation framework, and interface design. Although the final dashboard presents a clear decision workflow, the underlying process required repeated refinement in order to balance statistical quality, baseball interpretability, and practical usability. The final system did not emerge from a single modeling decision. Instead, it was the result of a sequence of adjustments in response to methodological limitations, conceptual mismatches, and product-design constraints.

12.1: Clustering and Model Selection Challenges

One of the first major challenges was that the early clustering approaches did not produce player groupings that were sufficiently meaningful for baseball decision-making. Initial methods such as k-means, Gaussian mixture models, and k-medoids were explored during model development, but these approaches often failed to generate clusters that were both statistically coherent and easy to interpret in

baseball terms. In some cases, players with visibly different offensive or defensive identities were placed together. In other cases, players who seemed to serve similar roster roles were split across clusters in ways that were difficult to justify conceptually.

A related challenge involved determining the appropriate level of cluster granularity. While larger values of clusters sometimes revealed more detailed player differences, they also produced overly narrow groups that were difficult to describe, compare, and use in a practical decision context. This created a tension between precision and usability. A model with many clusters may appear more detailed, but if those clusters are too fragmented, they become less useful for a front-office-style replacement tool. The eventual move to spectral clustering, and especially to the final 5D specification, addressed this issue well by producing broader archetypes that were both interpretable and statistically strong.

12.2: Defining Similarity and Adding Risk

Another important challenge was defining similarity in a way that matched the real baseball problem. A simple replacement system might rank players based only on WAR or other surface-level production statistics. However, two players can have similar overall value while contributing in very different ways. One shortstop may create value through speed, contact, and baserunning, while another may rely more heavily on power and defense. If the model ignored those differences, the resulting recommendations would often fail to reflect actual roster fit.

For that reason, the project moved away from one-dimensional similarity and toward an archetype-aware recommendation framework. The goal was not to identify statistically identical players in the narrowest possible sense, but to help users find replacements who either resemble the unavailable player closely or provide a comparable contribution through a different but still meaningful profile.

At the same time, similarity alone was not sufficient to produce realistic recommendations. Even when two players appeared close in profile space, they could differ sharply in age, injury exposure, or contract burden. To address this limitation, the team incorporated a separate risk framework built on age risk, injury risk, and contract risk. This moved the project beyond a purely descriptive matching model and toward a more practical decision framework, where player fit and acquisition feasibility are evaluated together.

12.3: Interface Simplicity and Usability

The project also faced a product-design challenge: the dashboard had to remain understandable to a non-technical user. Even a strong analytical model can fail in practice if its outputs are too difficult to interpret or if the interface is cluttered with technical detail. Since the final deliverable was an interactive replacement finder

rather than a notebook-style technical output, the team had to simplify the user experience without removing the most important information.

This required several rounds of revision to the layout, recommendation cards, filter structure, cluster labels, and explanatory notes. The final dashboard was organized around a practical workflow: identify the player to replace, review that player's profile and risk, inspect ranked alternatives, and then narrow the candidate set using intuitive filters such as age, contract level, cluster type, injury history, and total risk. These interface iterations were not just cosmetic. They were closely tied to the usefulness of the final product, since the dashboard had to translate a complex modeling process into an accessible decision tool.

12.4: Overall Reflection

Taken together, these challenges show that the final project was shaped by repeated refinement rather than linear execution. The team moved from weaker clustering methods to a stronger spectral-clustering framework, from broad feature experimentation to more interpretable dimensions, from simplistic similarity logic to archetype-aware matching, and from pure similarity scoring to a recommendation system adjusted by risk. At the same time, the interface was redesigned so that the final dashboard could communicate the model's outputs clearly to a non-technical audience. This iterative process was one of the most valuable outcomes of the project, because it demonstrated that a strong applied analytics tool requires not only technical accuracy, but also interpretability, domain awareness, and clear decision support.

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KEY TAKEAWAYS AND CONCLUSION

This project developed a usable replacement finder for 2025 MLB shortstops by combining clustering, risk modeling, and similarity-based recommendation into a single integrated framework. Rather than treating replacement as a simple ranking problem, the project approached it as a multidimensional decision task involving production, player type, availability, and contract feasibility.

Several important takeaways emerged from the final results. First, the project showed that WAR alone is not enough for realistic replacement analysis. Although WAR is a useful summary metric, it cannot fully capture how a player creates value, nor can it reflect practical constraints such as injury risk or contract burden. A strong replacement is not simply the player with the closest aggregate value. Instead, a

strong replacement is one whose overall profile makes sense in the context of team needs, roster structure, and acquisition feasibility.

Second, the results suggest that broad archetypes are more useful than highly fragmented clusters for this type of tool. The chosen spectral clustering model, Feature Set D with five clusters, produced interpretable shortstop types that aligned well with baseball intuition. These archetypes made it easier to distinguish among elite all-around players, speed-oriented shortstops, contact profiles, power/defense players, and balanced contributors. More importantly, they allowed the recommendation system to be grounded in role-based similarity rather than raw statistical closeness alone.

Third, the inclusion of risk made the recommendations substantially more realistic. Similarity by itself can identify interesting comparison players, but it cannot answer whether those players are practical targets. The incorporation of age, injury, and contract considerations made the tool more aligned with real decision-making, since teams must evaluate both fit and feasibility. In this framework, risk is not interpreted as a measure of talent. Rather, it functions as a measure of uncertainty and acquisition burden, which is a more appropriate role in replacement analysis.

Fourth, the final dashboard demonstrates the value of making complex analytics accessible. The tool is structured so that a user can review the selected player's profile, interpret the associated risks, view ranked alternatives, and apply filters without needing to understand the technical details of clustering or recommendation scoring. This accessibility is one of the strengths of the final product, because analytical insight is only useful when it can be clearly interpreted and applied.

Overall, the project accomplished its central objective. It produced a practical decision-support tool for identifying shortstop replacements in the 2025 player pool, while also showing that replacement evaluation is best treated as a structured and multidimensional problem. The final system combines clustering, risk, and similarity in a way that is both analytically grounded and operationally intuitive.

In conclusion, the MLB Shortstop Replacement Finder demonstrates that a stronger replacement framework can be built when player value is interpreted through archetypes, adjusted for risk, and delivered in an accessible interactive format. The project therefore contributes both a useful baseball analytics application and a broader example of how unsupervised learning, risk modeling, and product design can be combined into one coherent decision-support system.

14

FUTURE ROADMAP

Although the final tool performs well within the scope of this project, several clear extensions could improve both its realism and its broader applicability.

14.1: Extension to Other Positions

The current framework is limited to shortstops. This was a deliberate project choice, since shortstop offers a useful combination of offense, defense, and baserunning for archetype analysis. However, future work could extend the framework to other defensive positions. Such an extension would make the tool more broadly useful for roster planning and would test whether the same modeling structure performs equally well at positions with different value profiles.

This would be particularly interesting because replacement logic is position-dependent. Catchers, corner infielders, center fielders, and second basemen each create value in different ways, and the balance between offense, defense, and positional demands is not constant across the field. Expanding the framework would therefore increase both the scope and the realism of the overall system.

14.2: Including Minor League or Prospect Options

A second extension would be to incorporate minor league or near-major-league-ready players into the replacement pool. The current dashboard evaluates only MLB shortstops, which makes sense for a first version but excludes an important source of potential internal replacements.

In actual baseball operations, replacement decisions are not limited to external major league options. Teams often consider internal promotions, prospect timelines, and organizational depth when a player becomes unavailable. Adding minor league data would therefore make the model more realistic and strategically useful, especially for organizations deciding between external acquisition and internal development.

14.3: Adding Market Realism

The current framework accounts for risk and contract burden, but it does not explicitly model trade availability or market realism. A player may appear to be an ideal replacement according to similarity and risk, but still be unrealistic to acquire in practice because of team context, competitive incentives, contract structure, or expected trade cost.

A natural next step would be to add a market-realism layer to the model. This could include measures of trade likelihood, team competitiveness, roster surplus, or

approximate acquisition cost. Even relatively simple proxies for availability would improve the realism of the recommendations and make the dashboard more closely aligned with actual front-office constraints.

14.4: Improving the Injury Model

The injury-risk component already improves on simple injury counts by incorporating missed days, expected missed days, injury frequency, and severity. Still, injury modeling remains an area with significant room for improvement.

Future work could explore richer injury-history features such as time since last injury, recurrence patterns, injury type, body region, or role-specific durability effects. Because availability is often the most immediate concern in a replacement scenario, improvements to the injury model could have a substantial impact on the overall recommendation framework.

14.5: Updating the Tool With Future Seasons

Finally, the model should be updated as future seasons become available. The current dashboard uses 2023–2025 player-seasons for model development and the 2025 player pool for the final tool. While this is appropriate for the current version of the project, player roles, contracts, and risks change over time.

A future version of the dashboard could be updated annually, or even more frequently, so that the replacement recommendations remain aligned with the current state of the player pool. Doing so would also make it possible to study how archetypes evolve across seasons and whether the structure of the replacement market changes over time.

14.6: Final Note on Future Development

Taken together, these extensions suggest that the current project is a strong foundation rather than a closed endpoint. Extending to other positions, incorporating minor league options, modeling acquisition realism more directly, improving the injury framework, and updating the data over time would all make the system more robust and more realistic. These directions point toward a broader baseball decision-support platform built on the same core framework.

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REFERENCES

Baseball Savant. Statcast player data and leaderboards:

<https://baseballsavant.mlb.com/>

FanGraphs. RosterResource injury report:

<https://www.fangraphs.com/roster-resource/injury-report>

FanGraphs. RosterResource payroll and contract data:

<https://www.fangraphs.com/roster-resource/payroll/>

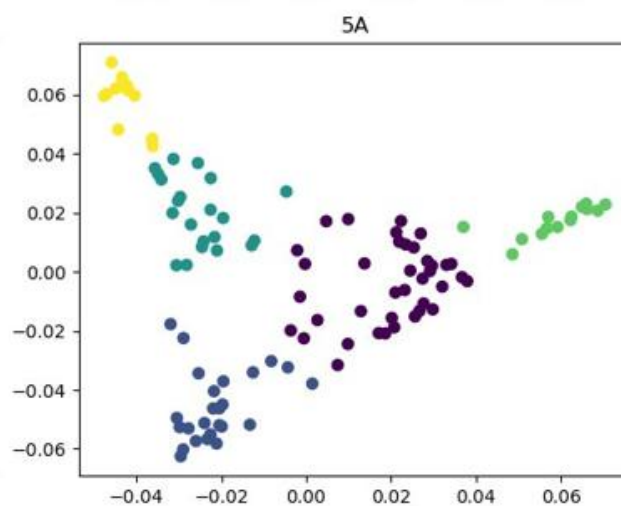
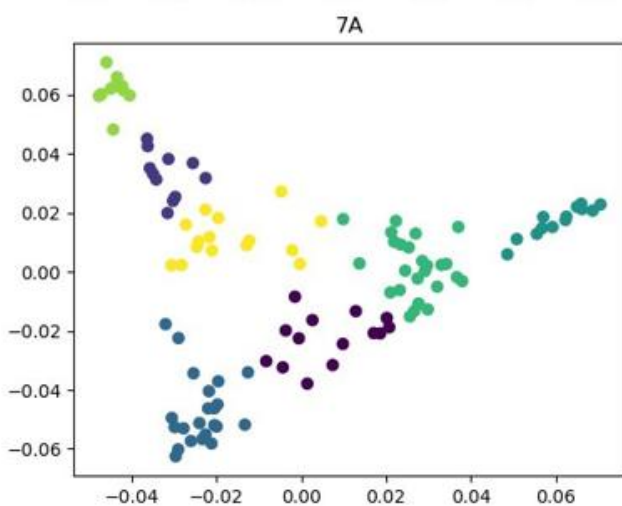
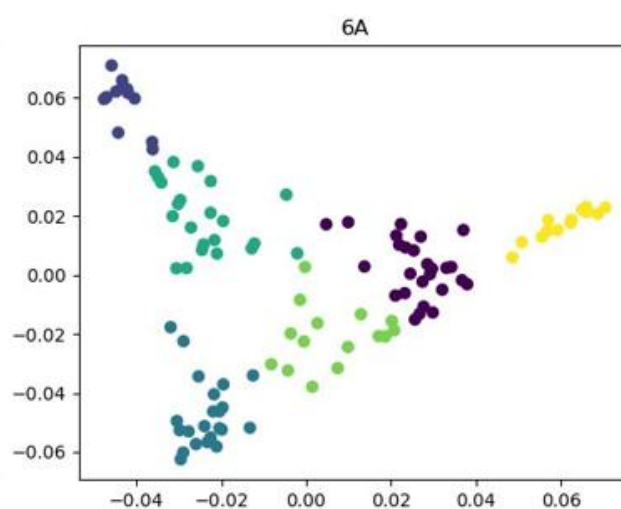
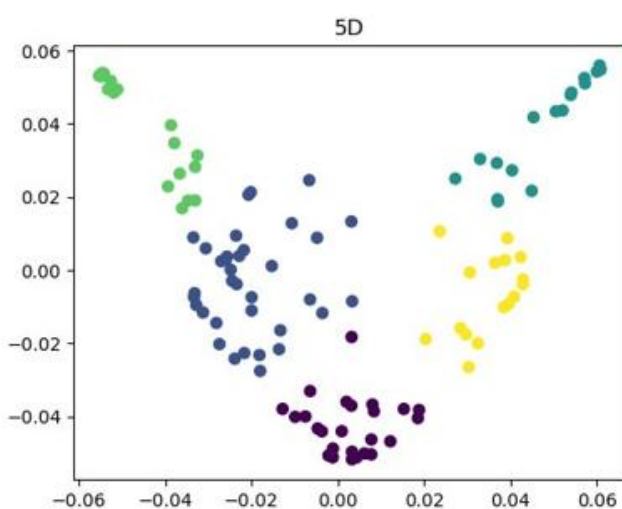
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APPENDIX

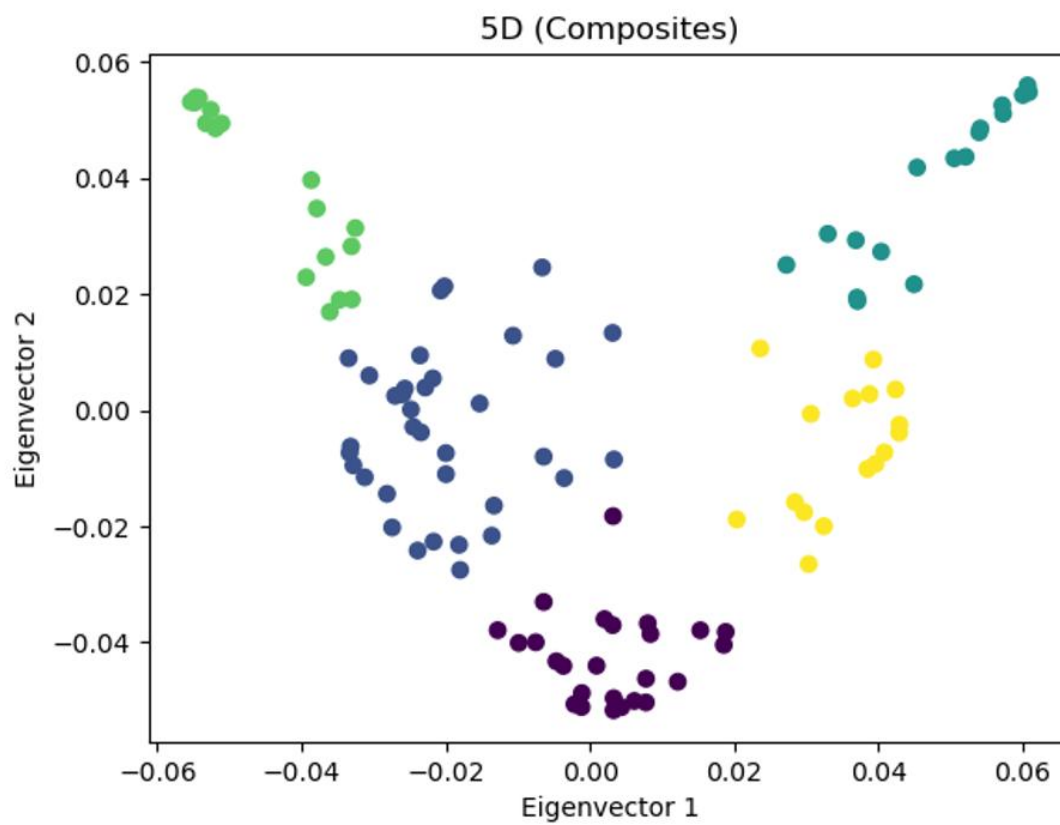
Appendix A: Clustering comparisons and eigenvector-space plots

Model	Silhouette Score	Best Feature Set
K-means	0.13-0.25	6D
Spectral clustering	0.50-0.53	5D
K-medoids	0.074-0.21	7D
GMM	0.12-0.16	7B

Feature set	Silhouette score	Dunn Index
5A	0.553	0.128
5D	0.532	0.154
6A	0.537	0.111
6D	0.502	0.0651
7A	0.531	0.116
7D	0.508	0.0659



Appendix B: Final model spectral-embedding clustering



Appendix C: Risk distribution

